

VOLTEDGE ELECTRONICS

Project Summary

Business context · data model · report structure

Fictional company · synthetic seeded data

01 — Business Context & KPI Framework

This document defines *the business* the analytics are built for: who VoltEdge Electronics is, how it makes money, who consumes the report, and the exact definition of every metric. Everything downstream (data model, DAX, report pages) traces back to this document.

1. Company profile

Attribute	Value
Name	VoltEdge Electronics
Type	Pure-play online retailer (direct-to-consumer), no physical stores
Category	Consumer electronics
Founded	2022 (US); trading history predates the analytics extract
HQ	Austin, TX, USA
Reporting currency	USD
Fulfillment	Company-operated fulfillment centers + regional 3PL partners
Analytics extract window	2024-07-01 → 2026-06-30 (24 months)

Why 24 months

Two complete trailing years lets every metric be shown **year-over-year against the same period last year**:

Period	Range
Current year (CY)	2025-07-01 → 2026-06-30
Prior year (PY)	2024-07-01 → 2025-06-30

Market expansion timeline

VoltEdge scaled from one market to four *during the extract window*. The report must make this legible — total revenue “steps up” as markets go live, and blended metrics (CAC, AOV, margin, delivery time) shift as the market mix changes.

Market	Currency	Live since	In the extract	Fulfillment centers	Primary carriers
United States	USD	2022	full 24 months	Dallas TX, Reno NV	UPS, FedEx, USPS
United Kingdom	GBP	2025-01-01	18 months	Birmingham	Royal Mail, DPD, Evri
Germany	EUR	2025-01-01	18 months	Leipzig	DHL, Hermes, DPD
Brazil	BRL	2025-07-01	12 months	São Paulo (Cajamar)	Correios, Jadlog, Loggi

Product catalog

~300 active SKUs organized as **Category** → **Subcategory** → **Brand** → **Product**.

Category	Example subcategories	Typical gross margin
Smartphones	Flagship, Mid-range, Budget	8–14%
Laptops & Tablets	Ultrabooks, Gaming laptops, Tablets	10–16%
Audio	Headphones, Earbuds, Speakers, Soundbars	22–35%
Gaming	Consoles, Controllers, VR, Games	6–18%
Smart Home	Hubs, Cameras, Lighting, Thermostats	20–30%
Wearables	Smartwatches, Fitness bands	18–28%
Accessories	Cables, Chargers, Cases, Storage, Power banks	35–55%

Why this matters analytically: electronics is a **thin-margin** business. Flagship phones and consoles drive revenue but barely any profit; accessories and audio carry the margin. A recurring executive question is “*are we growing revenue or growing profit?*” — the answer depends on **category mix**, which is why price–volume–mix analysis is a first-class citizen in this report.

2. Business model & economics

Revenue streams

- 1. **Product sales** — the vast majority of revenue.
- 2. **Shipping fees** — charged on orders below a free-shipping threshold (varies by market).
- 3. **Extended warranty / protection plans** — attach-rate add-on, high margin, modeled as a separate line item (`line_type = 'warranty'`) on a share of eligible orders.

Cost structure (modeled)

- **COGS** — unit cost × quantity (per SKU, sourced from the purchase price of the stock actually sold).
- **Discounts** — promotional and coupon reductions (campaign-driven and site-wide sales).
- **Returns** — refunded merchandise value (electronics return rates are high: 8–14%).
- **Marketing spend** — paid media across Paid Search, Paid Social, Display, Affiliate, plus email tooling.
- **Shipping cost** — what VoltEdge pays carriers (distinct from shipping *fees* charged to customers).
- **Payment processing** — merchant fees per transaction + financing cost of merchant-funded installments (\$3).
- **Inventory purchases** — cash paid to suppliers for stock, on supplier payment terms (\$4).

Financial definitions used throughout

Term	Definition
Gross Revenue (GMV)	Σ (unit list price × quantity), before discounts, excl. tax & shipping
Discounts	Σ discount amount applied to order lines
Returns	Σ net merchandise value of returned lines (post-discount)
Net Revenue	Gross Revenue – Discounts – Returns. <i>This is the North Star.</i> Excludes tax and shipping fees.
COGS	Σ (unit cost × net quantity sold)
Gross Profit	Net Revenue – COGS
Payment Fees	Merchant discount rate × amount + fixed fee per transaction (+ merchant-funded financing cost)
Contribution Margin	Gross Profit – Marketing Spend – Shipping Cost – Payment Fees (operating-contribution proxy; not a full P&L)

Net Revenue, COGS and margin are always expressed in **USD (reporting currency)**. Transactions are captured in **local currency** and converted using the exchange rate on the transaction date (see §6).

3. Payment methods, fees & customer-side cash

A deliberate CFO-grade thread. Payment choice affects **cost** (processing fees differ a lot by scheme) and **cash timing** (settlement lag, and Brazilian installments spreading cash over months).

`dim_payment_method` attributes

Attribute	Meaning
<code>method_group</code>	Card · Digital Wallet · Bank Transfer · BNPL · Pix · Boleto
<code>method_name</code>	Visa Credit, Mastercard Debit, Amex, Elo Credit, Hipercard, PayPal, Apple Pay, Google Pay, Klarna, Clearpay, SEPA Direct Debit, Pix, Boleto Bancário
<code>card_scheme</code>	Visa · Mastercard · Amex · Elo · Hipercard · (n/a)
<code>mdr_pct</code>	Merchant discount rate (% of transaction)
<code>fixed_fee_local</code>	Flat fee per transaction
<code>settlement_days</code>	Days until VoltEdge receives the cash
<code>installment_eligible</code>	Supports paying in instalments
<code>max_installments</code>	Cap (12 in BR, 3–4 for EU BNPL)
<code>available_markets</code>	Methods are market-specific (Pix/Boleto/Elo/Hipercard = BR only; SEPA/Klarna/Clearpay = UK/DE; Amex skewed US/UK)

Indicative economics (modeled, not a real rate card)

Method	<code>mdr_pct</code>	Fixed fee	Settlement days	Notes
Visa / Mastercard – Debit	1.1 %	—	2	cheapest card
Visa / Mastercard – Credit	2.2 %	—	2	
Amex	3.4 %	—	3	highest scheme fee
Elo / Hipercard (BR) – Credit	2.6 %	—	2 (à vista)	instalments settle monthly
PayPal	2.9 %	0.30	1	
Apple Pay / Google Pay	2.0 %	—	2	rides card rails
Klarna / Clearpay (BNPL)	4.5 %	0.30	2	merchant paid upfront, keeps the fee
SEPA Direct Debit (DE)	0.8 %	0.25	3	
Pix (BR)	0.45 %	—	0	instant, cheapest overall
Boleto Bancário (BR)	—	3.49	1 (after customer pays, avg +2)	~12 % of boletos are never paid → order cancelled

Installments (`installment_plan`)

Plan	Who pays financing	Cash-flow effect
<code>None</code> (à vista)	—	Full amount settles once, after <code>settlement_days</code>
<code>Interest-Free</code> (merchant-funded — BR "parcelado sem juros")	VoltEdge absorbs ~1.8 %/month of the deferred balance	Cash arrives in <i>N</i> monthly tranches; a financing cost is booked
<code>With Interest</code> (customer-funded)	Customer	Cash arrives in <i>N</i> monthly tranches; no extra merchant cost

- `fact_orders` carries the order-grain payment attributes and total `payment_fee`.
- `fact_payment_schedule` explodes each order into **one row per settlement tranche** (`installment_number` , `due_date` , `gross_amount` , `fee_amount` , `financing_cost` , `net_amount` , currency) — the table the customer-side cash-flow visuals sit on.

4. Inventory, procurement & the cash conversion cycle

The other half of working capital: cash goes **out** to suppliers to buy stock, that stock **sits** in a warehouse, then a sale converts it back to cash **in**. The report treats this as a first-class CFO/COO topic.

What is modeled

Object	Grain	Purpose
dim_supplier	supplier	Country, payment_terms_days (Net 30 / 45 / 60), lead_time_days , on-time-supply reliability
fact_purchase_orders	PO line (PO × SKU)	Order date, supplier, qty ordered, unit cost, PO value (local + USD), expected & actual receipt date, supplier payment due date (receipt + terms), supplier paid date
fact_inventory_movement	movement	Transactional ledger: receipt (from a PO), sale (from an order line), return_to_stock , adjustment , transfer . Date, SKU, warehouse, qty (+/-), unit cost, value
fact_inventory_snapshot	month-end × SKU × warehouse	Derived from the movement ledger. Units on hand, units in transit (inbound), units reserved, unit cost, inventory value (local + USD), days of cover

Building the snapshot **from** the movement ledger is a deliberate data-engineering showcase (a running balance / window function per SKU-warehouse), documented in [04_etl_pipeline.md](#) .

COGS sourcing

COGS on a sale is taken from the **weighted-average purchase cost** of stock on hand for that SKU at sale time (moving-average cost), so margin reflects what the goods actually cost — not a static list cost. Price drift in fact_purchase_orders therefore flows through to margin.

Working-capital KPIs (Executive Summary tile + dedicated Phase-2 section)

KPI	Definition	Benchmark
Inventory Value (on hand)	Σ units on hand × moving-avg cost, USD	—
Inventory in Transit	Σ inbound units not yet received × PO cost	—
DIO — Days Inventory Outstanding	Avg Inventory Value ÷ COGS × days in period	45–75
DSO — Days Sales Outstanding	Amount-weighted avg days from order date to cash settlement	US/EU ~2–4; BR much higher (instalments)
DPO — Days Payables Outstanding	Avg Accounts Payable ÷ COGS (or purchases) × days in period	30–60
CCC — Cash Conversion Cycle	DIO + DSO – DPO	lower is better; watch the trend
Accounts Payable (open)	Σ received-but-unpaid PO value at a date	—
Accounts Receivable (open)	Σ booked-but-unsettled customer amounts at a date	—
Working Capital Tied Up	Inventory Value + AR – AP	—
Supplier Spend	Σ PO value received in period, USD	—
Inventory Turnover	Trailing COGS ÷ Avg Inventory Value	5–8×/yr
Weeks of Cover	Inventory on hand ÷ avg weekly units sold	4–10
Stockout Rate %	SKU-days out of stock ÷ SKU-days	< 3%
Aged / Excess Inventory %	Value with > 90 days of cover ÷ total inventory value	< 15%

Company cash-flow timeline

A single view combining, per week/month:

- **Cash out** — supplier payments (PO value on supplier_paid_date), marketing spend, shipping cost, payment processing fees.
- **Cash in** — customer settlements from fact_payment_schedule .
- **Net cash flow** and **cumulative cash position**, plus the split of working capital between inventory and receivables. Makes the cost of BR "parcelado sem juros" and of slow-moving stock visible in cash terms.

5. Total vs Comparable Base (like-for-like)

Because markets go live mid-window, a raw YoY number mixes *organic* growth with *expansion* growth. Every headline growth metric is shown **two ways**:

View	Definition
Total	All markets, as reported.
Comparable Base	Only markets live for the entire CY + PY span → United States only. The like-for-like / "same-store" view.
Expansion Contribution	Total – Comparable Base. Revenue / orders / customers attributable to UK, DE, BR.

Secondary cut: **Core-3 Comparable** (US + UK + DE, comparable from 2025-01) for analysis from 2025 onward.

Implementation: `dim_market` carries `live_date` ; a calculated `is_comparable_base` (relative to PY start 2024-07-01) flags US. DAX also provides a **dynamic** comparable-base measure that recomputes the eligible market set from the visual's date context. Both are in [06_dax_measures.md](#) .

6. Multi-currency handling

- Every monetary transaction is stored in its **local currency** with a `currency_code` .
- `fact_exchange_rate` holds a **daily** rate per currency → USD.
- Curated fact tables carry **both** `amount_local` and a materialized `amount_usd` (converted at transaction-date rate) so results are reproducible without the model.
- DAX also exposes **currency-aware measures** and a **reporting-currency selector** (USD / EUR / GBP / BRL). Default = USD.
- FX approach: rates fluctuate around anchors (EUR 1.08, GBP 1.26, BRL 0.19 USD per unit) with realistic daily drift.

7. Seasonality & patterns embedded in the data

- **Q4 peak** — Nov/Dec ~1.7–2.1× a baseline month; **Black Friday / Cyber Monday** spike within November.
- **Secondary peaks** — mid-year sale (July), back-to-school (late Aug / Sep).
- **Growth trend** — ~2–4% MoM growth on the comparable base, plus step changes at each market launch.
- **Channel mix evolution** — heavy Paid Search/Social early; Organic, Email, Direct grow as brand awareness builds; CAC rises over time.
- **Category trends** — Wearables and Smart Home grow share; Smartphones' share slowly declines.
- **Purchase-cost drift** — supplier unit costs move over time; some SKUs discontinued and replaced → margin and inventory-aging effects.
- **Promo periods** — recurring discount events with elevated discount rates and order volume.
- **Payment-mix shift** — Pix share in BR grows MoM; BNPL share in UK/DE grows; Amex share slowly falls.
- **Fulfillment stress** — delivery times and late-delivery rates worsen during Q4 and vary by carrier and destination region.
- **Returns** — higher for Smartphones/Laptops, lower for Accessories; split "defective" vs "changed mind" / "not as described".

8. Stakeholders & the questions they ask

The report is **executive-level**: each page answers the standing questions of one leader.

Persona	Owns	Standing questions
CEO	Whole business	Are we hitting plan? Which market/category drives or drags growth? Revenue <i>and</i> margin trajectory? Total vs like-for-like?
CFO	Margin, cash & efficiency	Gross margin trend and why it moved. Discount leakage. Payment fees and the cost of "sem juros". Cash received vs revenue booked. Cash conversion cycle — how much cash is tied up in inventory and receivables? Contribution margin by market. Revenue vs target.
CMO	Demand & acquisition	CAC and ROAS by channel and campaign. New vs returning revenue. Which campaigns paid back? Funnel drop-off.
VP E-commerce	Site & conversion	Traffic, conversion rate, AOV, cart abandonment, revenue per session, device performance.
COO / VP Operations	Fulfillment & inventory	On-time delivery, delivery cycle time, shipping cost per order, carrier performance, stockouts and weeks of cover, aged inventory , inventory turnover, supplier lead time & reliability.
VP Customer	Retention & service	Repeat rate, cohort retention, CLV, churn, support volume & resolution time, CSAT, return rate by reason.

9. KPI framework

Metrics defined in business terms here; exact DAX in [06_dax_measures.md](#) . "Benchmark" is a plausible DTC-electronics reference range used to set report targets — not a claim about any real company.

9.0 North Star

KPI	Definition	Benchmark
Net Revenue (USD)	Gross Revenue – Discounts – Returns, converted to USD	vs monthly plan

9.1 Sales & Growth (Executive Summary, Sales Performance)

KPI	Definition	Benchmark
Gross Revenue (GMV)	$\Sigma (\text{list price} \times \text{qty})$	—
Net Revenue	Gross Revenue – Discounts – Returns	vs target
Orders	Distinct orders, excluding fully cancelled	—

KPI	Definition	Benchmark
Units Sold	Σ net quantity	—
Average Order Value (AOV)	$\text{Net Revenue} \div \text{Orders}$	\$170–\$260
Units per Order	$\text{Units} \div \text{Orders}$	1.6–2.2
Gross Profit	$\text{Net Revenue} - \text{COGS}$	—
Gross Margin %	$\text{Gross Profit} \div \text{Net Revenue}$	12–20% blended (thin, by design — see note below)
Discount Rate %	$\text{Discounts} \div \text{Gross Revenue}$	4–10%
Return Rate %	$\text{Returned Units} \div \text{Units Sold}$	7–12%
Warranty Attach Rate %	$\text{Orders with a warranty line} \div \text{eligible orders}$	6–12%
Net Revenue YoY % — Total	$(\text{CY} - \text{PY}) \div \text{PY}$, all markets	—
Net Revenue YoY % — Comparable Base	same, US-only like-for-like	—
Expansion Contribution	$\text{Total Net Revenue} - \text{Comparable-Base Net Revenue}$	—
Revenue vs Target %	$\text{Net Revenue} \div \text{Target} - 1$	≥ 0
Price / Volume / Mix	Decomposition of Net Revenue Δ vs PY into rate, quantity, category-mix effects	—
Revenue by Market / Category / Channel	Net Revenue sliced by dimension	—

Thin margin is the story, not a bug. Blended gross margin sits at ~14% — flagship phones, laptops and consoles carry almost no margin; audio and accessories carry it all. After marketing (~\$4M), shipping and payment fees, **contribution margin is near break-even** — VoltEdge is a fast-growing, not-yet-profitable business. The report is built to make that tension legible: *are we buying growth we can't afford, and does category mix / repeat revenue get us to profit?*

9.2 Payments & Customer Cash (Executive Summary, Sales Performance > Payments)

Payment Processing Cost · Effective Fee Rate % · Fee Rate by Scheme/Method/Market · Merchant-Funded Financing Cost · Installment Mix % · Avg Instalments (BR) · Revenue Booked vs Cash Received · Accounts Receivable (open) · DSO. (Definitions in §3.)

9.3 Working Capital & Inventory (Executive Summary tile; dedicated Phase-2 section)

DIO · DSO · DPO · CCC · Inventory Value · Inventory in Transit · Accounts Payable (open) · Accounts Receivable (open) · Working Capital Tied Up · Supplier Spend · Inventory Turnover · Weeks of Cover · Stockout Rate % · Aged / Excess Inventory % · Company cash-flow timeline. (Definitions in §4.)

9.4 Marketing & Acquisition (Marketing & Acquisition)

KPI	Definition	Benchmark
Marketing Spend	Σ paid media spend (+ email tooling)	—
Impressions / Clicks	From ad-platform exports	—
CTR %	$\text{Clicks} \div \text{Impressions}$	0.7–2.5%
CPC	$\text{Spend} \div \text{Clicks}$	\$0.40–\$1.60
New Customers	Customers whose first-ever order falls in the period	—
CAC (paid)	$\text{Paid Marketing Spend} \div \text{New Customers via paid channels}$	\$45–\$110
Blended CAC	$\text{Total Marketing Spend} \div \text{Total New Customers}$	\$30–\$70
ROAS	$\text{Attributed Net Revenue} \div \text{Ad Spend (last-touch, channel-level)}$	3.0–6.0×
MER	$\text{Total Net Revenue} \div \text{Total Marketing Spend}$	6–10×
New vs Returning Revenue	Net Revenue split by customer order sequence	—
Campaign Performance	Spend, Orders, Net Revenue, ROAS per campaign	ROAS $\geq 3\times$ to pay back
Acquisition Funnel	Impressions → Clicks → Sessions → Add-to-Cart → Orders	—

9.5 Website / Digital (Phase 2)

Sessions / Users · Conversion Rate % (Orders $\div$ Sessions, 1.3–2.6%) · Bounce Rate % · Pages per Session · Add-to-Cart Rate % · Cart Abandonment Rate % · Revenue per Session · Device / Browser mix.

9.6 Logistics & Fulfillment (Phase 2)

Avg Delivery Time (days, 3–6) · On-Time Delivery % (92–96%) · Perfect Order Rate % (88–93%) · Ship Time (days) · Shipping Cost per Order (\$6–\$14) · Shipping Cost Recovery % · Carrier Scorecard · Supplier Lead Time & On-Time-Supply % · (inventory metrics in §9.3).

9.7 CRM / Customer (Phase 3)

Active Customers · New / Returning · Repeat Purchase Rate % (25–35%) · Cohort Retention % · Churn Rate % · CLV · RFM Segment · Loyalty Tier Mix · Support Tickets · Avg Resolution Time (h, <24) · First Contact Resolution % (70–80%) · CSAT (≥4.3) · Return Rate by Reason.

10. Targets / plan

`fact_target` holds a **monthly plan by market** for: Net Revenue, Orders, New Customers, Blended CAC, Gross Margin %. Targets are "ambitious but plausible" (actuals land ~90–110% of plan). Targets exist only for months a market is live.

11. Out of scope (explicit)

- Full P&L / GAAP accounting, tax filing detail, statutory cash-flow statement.
- Real ad-platform attribution modeling (simple last-touch at channel level).
- Employee, payroll, or fixed-cost data.
- Real personal data — all customers synthetic (`Faker` , seeded).
- Real-time / streaming — batch pipeline, one historical load.

12. Resolved assumptions

#	Assumption	Decision
1	North Star = Net Revenue (excl. tax & shipping fees), USD; Gross Profit secondary	✓
2	Return rate ~8%, blended gross margin ~14% (thin by design; growth-stage, near break-even after marketing)	✓
3	Extract window 24 months : 2024-07-01 → 2026-06-30	✓
4	Expansion inside window: US always-on · UK+DE 2025-01-01 · BR 2025-07-01	✓
5	Every growth metric shown Total and Comparable Base (US like-for-like) + Expansion Contribution	✓
6	Warranty / protection-plan revenue modeled as an add-on line	✓
7	Reporting-currency selector (USD/EUR/GBP/BRL) in the model	✓
8	Payment model: MDR by scheme, fixed fees, settlement lag, instalments (merchant- vs customer-funded), <code>fact_payment_schedule</code>	✓
9	Procurement & inventory modeled: <code>dim_supplier</code> (Net 30/45/60), <code>fact_purchase_orders</code> , <code>fact_inventory_movement</code> ledger → derived <code>fact_inventory_snapshot</code> ; moving-average COGS; DIO/DSO/DPO/CCC and a company cash-flow timeline	✓
10	Realised scale: 294 SKUs · ~103k customers · ~158k non-cancelled orders · ~262k order lines · ~14k PO lines · 24 months. AOV ~\$208, ~1.7 lifetime orders/customer, repeat rate ~40%.	✓

05 — Data Model (Curated Star Schema)

The curated layer is a Kimball-style star: one row per business event in the fact tables, conformed dimensions with integer surrogate keys, a real date dimension, and every monetary value available in both local currency and USD.

Power BI connects to `data/curated/*.parquet` through a single parameter (`pDataFolder`).

1. Bus matrix — which dimension touches which fact

Dimension ▾ / Fact ▾	date	market	customer	product	supplier	channel	campaign	payment_method	carrier	warehouse	currency
fact_order_lines	• (order/ship/deliver)	•	•	•		•		•		•	•
fact_orders	• (order/ship/deliver)	•	•			•		•	•	•	•
fact_payment_schedule	• (order/due)	•									•
fact_returns	• (return)	•	•	•							•
fact_purchase_orders	• (order/receipt/due/paid)	•		•	•					•	
fact_inventory_movement	• (movement)	•		•						•	
fact_inventory_snapshot	• (month-end)	•		•						•	
fact_marketing_spend	•	•				•	•				
fact_web_traffic_daily	•	•				•					
fact_support_tickets	• (created/resolved)	•	•								
fact_exchange_rate	•										•
fact_target	• (month)	•									

All date roles join to the **single** `dim_date` (Power BI: one active relationship on `order_date_key` ; `ship_date_key` , `delivery_date_key` , etc. are inactive relationships activated with `USERELATIONSHIP` inside specific measures).

2. Fact tables — grain & measures

fact_order_lines (centre of the model)

Grain: one row per order line — an order × product, or a `warranty` add-on line. **Keys:** `order_date_key` , `ship_date_key` , `delivery_date_key` , `product_key` , `customer_key` , `market_key` , `currency_key` , `channel_key` , `warehouse_key` , `payment_method_key` ; degenerate `order_id` , `order_line_id` , `line_type` , `order_status` . **Measures:** `quantity` , `unit_price_local` , `gross_amount_(local|usd)` , `discount_amount_(local|usd)` , `net_amount_(local|usd)` , `cogs_usd` (moving-average), `gross_profit_usd` , `returned_qty` , `is_returned` , `refund_amount_(local|usd)` .

fact_orders

Grain: one row per order header. **Keys:** `order/ship/delivery date` , `customer_key` , `market_key` , `currency_key` , `channel_key` , `warehouse_key` , `carrier_key` , `payment_method_key` ; degenerate `order_id` , `order_status` , `device` , `installments` , `installment_plan` . **Measures:** `units` , `gross/discount/net/tax/shipping_fee/payment_fee` in `_local` and `_usd` , `shipping_cost_usd` , `ship_days` , `delivery_days` ; flags `is_late` , `is_on_time` , `is_delivered` , `is_cancelled` , `has_return` , `is_perfect_order` .

fact_payment_schedule

Grain: one row per settlement tranche (`order_id` × `installment_number`). **Keys:** `order_date_key` , `due_date_key` , `market_key` , `currency_key` . **Measures:** `n_installments` , `installment_plan` , `settlement_lag_days` , `gross/fee/financing_cost/net amount` in `_local` and `_usd` . Cash **received** in a period = SUM(`net_amount_usd`) filtered by `due_date_key` ; revenue **booked** = `fact_order_lines` filtered by `order_date_key` . The gap is the DSO story.

fact_returns

Grain: one return line. `return_date_key` , `product_key` , `customer_key` , `market_key` , `currency_key` ; `returned_qty` , `refund_amount_(local|usd)` , `reason` , `restocked` , `condition` .

fact_purchase_orders

Grain: one PO line (`po_number` × `SKU`). `order_date_key` , `expected/actual_receipt_date_key` , `payment_due_date_key` , `supplier_paid_date_key` , `supplier_key` , `product_key` , `warehouse_key` , `market_key` . **Measures:** `qty_ordered` , `unit_cost_usd` , `po_value_usd` , `payment_terms_days` ,

`late_receipt_days`, `days_to_pay`. Accounts payable open at a date = POs with `actual_receipt ≤ date < supplier_paid`.

fact_inventory_movement

Grain: one stock movement. `movement_date_key`, `product_key`, `warehouse_key`, `market_key`; `movement_type` (`adjustment` / `receipt` / `sale`), signed `quantity`, `unit_cost_usd` (moving-average at the date), `value_usd`.

fact_inventory_snapshot (derived from the ledger)

Grain: month-end × SKU × warehouse. `snapshot_date_key`, `month_year`, `product_key`, `warehouse_key`, `market_key`. **Measures:** `units_on_hand`, `moving_avg_cost_usd`, `inventory_value_usd`, `units_in_transit`, `avg_weekly_units_sold`, `weeks_of_cover`, plus `raw_snapshot_units` and `snapshot_variance_units` for the DQ narrative.

fact_marketing_spend

Grain: date × market × channel × campaign. `impressions`, `clicks`, `cost_local`, `cost_usd`.

fact_web_traffic_daily

Grain: date × market × channel × device. `sessions`, `users`, `pageviews`, `bounces`, `add_to_carts`, `carts_created`, `transactions`, `had_data_anomaly`.

fact_support_tickets

Grain: one ticket. `created_date_key`, `resolved_date_key`, `market_key`, `customer_key`, `category`, `priority`, `resolution_hours`, `first_contact_resolution`, `csat`.

fact_exchange_rate

Grain: date × currency. `rate_to_usd`, `rate_from_usd`. Used by currency-aware measures and the reporting-currency selector.

fact_target

Grain: month × market × metric. `target_value`. `metric` ∈ {Net Revenue, Orders, New Customers, Blended CAC, Gross Margin %}.

3. Dimensions — notable attributes

Dimension	Key	Attributes used in the report
dim_date	<code>date_key</code> (yyyymmdd)	<code>date</code> , <code>month_year</code> , <code>quarter_name</code> , <code>year</code> , <code>day_name</code> , <code>is_weekend</code> , <code>fiscal_year</code> / <code>fiscal_quarter</code> / <code>fiscal_period</code> (FY starts 1 Jul), <code>is_promo_period</code> , <code>promo_name</code> , <code>is_black_friday_week</code> , <code>in_extract_window</code> , <code>yoy_period</code> (CY/PY)
dim_market	<code>market_key</code>	<code>market_id</code> , <code>market_name</code> , <code>region</code> , <code>currency_code</code> , <code>live_date</code> , <code>is_comparable_base</code>
dim_currency	<code>currency_key</code>	<code>currency_code</code> , <code>currency_name</code> , <code>currency_symbol</code> , <code>is_reporting_currency</code>
dim_customer	<code>customer_key</code> (-1 = Unknown)	<code>market_id</code> , <code>acquisition_channel</code> , <code>acquisition_month</code> , <code>acquired_before_window</code> , <code>segment</code> , <code>age_band</code> , <code>loyalty_tier</code> , <code>marketing_consent</code> , <code>has_email</code> , <code>country</code> , <code>state_region</code> , <code>city</code> , <code>first_order_date</code> , <code>last_order_date</code> , <code>lifetime_orders</code> , <code>lifetime_net_revenue_usd</code> , <code>is_repeat_customer</code> , <code>customer_status</code> (Prospect/Active/Lapsed/Churned)
dim_product	<code>product_key</code> (-1 = Unknown/late-arriving)	<code>sku_id</code> , <code>product_name</code> , <code>category</code> , <code>subcategory</code> , <code>brand</code> , <code>list_price_usd</code> , <code>price_tier</code> , <code>launch_date</code> , <code>discontinue_date</code> , <code>is_active_at_window_end</code> , <code>is_warranty_eligible</code> , <code>warranty_months</code> , <code>weight_kg</code> , <code>supplier_key</code>
dim_supplier	<code>supplier_key</code>	<code>supplier_name</code> , <code>country</code> , <code>payment_terms_days</code> , <code>payment_terms_band</code> (Net 30/45/60), <code>lead_time_days</code> , <code>on_time_rate</code> , <code>reliability_band</code>
dim_channel	<code>channel_key</code>	<code>channel_name</code> , <code>channel_group</code> (Paid/Owned/Earned), <code>is_paid</code>
dim_campaign	<code>campaign_key</code>	<code>campaign_name</code> , <code>market_id</code> , <code>channel</code> , <code>campaign_theme</code>
dim_payment_method	<code>payment_method_key</code> (-1)	<code>method_name</code> , <code>method_group</code> , <code>card_scheme</code> , <code>mdr_pct</code> , <code>fixed_fee_local_nominal</code> , <code>settlement_days</code> , <code>installment_eligible</code> , <code>max_installments</code> , <code>available_markets</code>
dim_carrier	<code>carrier_key</code>	<code>carrier_name</code> , <code>market_id</code> , <code>baseline_transit_days</code> , <code>baseline_on_time_rate</code>
dim_warehouse	<code>warehouse_key</code>	<code>warehouse_id</code> , <code>warehouse_name</code> , <code>market_id</code> , <code>region</code>
small dims		<code>dim_device</code> , <code>dim_return_reason</code> , <code>dim_ticket_category</code> , <code>dim_order_status</code> , <code>dim_metric</code>

4. Cross-cutting model decisions

Multi-currency. Transactions are captured in local currency (`*_local`) with a `currency_key`, and every one carries a materialised `*_usd` at the order-date rate. `fact_exchange_rate` additionally lets DAX convert to any reporting currency on the fly; a disconnected `Reporting Currency` slicer drives

that. Default view is USD.

Total vs Comparable Base. `dim_market.is_comparable_base` (US only) gives the static like-for-like cut. A dynamic DAX version recomputes the eligible market set from the visual's date range, so the concept survives any filter. $\text{Expansion Contribution} = \text{Total} - \text{Comparable Base}$.

Moving-average COGS. `cogs_usd` on each sale line reflects the weighted-average purchase cost of that SKU at sale time (rebuilt from PO receipts), so margin moves with real sourcing cost, not a static list cost.

Inventory snapshot is derived, not trusted. `fact_inventory_snapshot` is rebuilt from the movement ledger; the raw WMS snapshot is kept only as `raw_snapshot_units` / `snapshot_variance_units` to quantify system drift.

Working-capital timing. Three clocks: goods received → paid supplier (`fact_purchase_orders`), stock held (`fact_inventory_snapshot`), order booked → cash settled (`fact_payment_schedule`). Together they give DIO, DPO, DSO and the cash conversion cycle, and a single company cash-flow timeline.

Unknown members. `product_key = -1` and `customer_key = -1` stubs absorb late-arriving / missing natural keys so fact rows are never dropped and totals always tie.

5. Power BI relationships (all single-direction, many-to-one from fact to dim)

<code>dim_date[date_key]</code>	<code>1 →* fact_order_lines[order_date_key]</code>	(active)
<code>dim_date[date_key]</code>	<code>1 →* fact_order_lines[ship_date_key]</code>	(inactive)
<code>dim_date[date_key]</code>	<code>1 →* fact_order_lines[delivery_date_key]</code>	(inactive)
<code>dim_market[market_key]</code>	<code>1 →* fact_order_lines[market_key]</code>	
<code>dim_customer[customer_key]</code>	<code>1 →* fact_order_lines[customer_key]</code>	
<code>dim_product[product_key]</code>	<code>1 →* fact_order_lines[product_key]</code>	
<code>dim_channel[channel_key]</code>	<code>1 →* fact_order_lines[channel_key]</code>	
<code>dim_currency[currency_key]</code>	<code>1 →* fact_order_lines[currency_key]</code>	
<code>dim_payment_method[...key]</code>	<code>1 →* fact_orders[payment_method_key]</code>	
<code>dim_carrier[carrier_key]</code>	<code>1 →* fact_orders[carrier_key]</code>	
<code>dim_warehouse[warehouse_key]</code>	<code>1 →* fact_orders[warehouse_key]</code>	, <code>fact_purchase_orders</code> , <code>fact_inventory_*</code>
<code>dim_supplier[supplier_key]</code>	<code>1 →* fact_purchase_orders[supplier_key]</code>	
<code>dim_campaign[campaign_key]</code>	<code>1 →* fact_marketing_spend[campaign_key]</code>	
<code>dim_date[date_key]</code>	<code>1 →* every other fact's primary date key</code>	

`fact_orders` and `fact_order_lines` are **not** related to each other — they are two grains of the same process and are sliced by shared dimensions, never joined.

VoltEdge Electronics — Report Structure

Document status: built to match the numbered `docs/` convention used by the other four portfolio projects. Full design rationale, palette, and page-by-page detail already live in [07_report_guide.md](#) — this doc is the compact index of that same report, table-first. One command rebuilds the report:

```
cd powerbi && bash build_report.sh
```

Pipeline: `gen_logo.py` → `gen_semantic_model.py` (+ `pbi_measures.py`) → `gen_dax_doc.py` → `gen_report_visuals.py` → `build_report.sh` (drives `pbir`). Output: `powerbi/VoltEdge_Electronics.pbip` + `.Report` + `.SemanticModel`. `pbir validate --all` passes (see the known polish backlog in `07_report_guide.md`).

First open in Power BI Desktop: Transform data → set `pDataFolder` to `data/curated` → Close & Apply · View ▸ Themes ▸ Browse → `powerbi/theme/VoltEdge.json`.

Semantic model

28 data tables (16 dims + 12 facts) + `_Measures` (167 measures in 14 folders) + two disconnected helper tables (`Reporting_Currency` , `Funnel_Stage`). 65 relationships, single-direction many-to-one fact → dimension; secondary date roles (ship, delivery, due, receipt, paid...) are inactive, activated per-measure with `USERELATIONSHIP`. See `docs/05_data_model.md` for the full bus matrix.

The seven pages

Every page: navy logo bar + accent rule (header image) · Currency + Market slicer row · Analysis-period date slicer · 6 KPI tiles · two content rows on the 3-30-300 gradient.

#	Page	Question	KPI row (6 cards)	Content
1	Executive Summary	Are we hitting plan, is growth real, where does the money go?	Net Revenue · Gross Margin % · Orders · Contribution Margin · New Customers · Blended CAC	Net revenue by month CY vs PY (line) · budget scorecard (table) · contribution-margin bridge (waterfall) · net revenue by market CY vs PY (column)
2	Sales Performance	What is driving the top line — volume, price, or mix?	Net Revenue · Orders · AOV · Gross Margin % · Discount Rate % · Units/Order	Net revenue & discount rate by month (combo) · net revenue by category (bar) · YoY revenue change by category (waterfall, price/volume/mix) · cumulative revenue booked vs cash received (line)
3	Marketing & Acquisition	Is paid spend efficient, and how much growth is actually paid-for?	Marketing Spend · Blended CAC · MER · New Customers · % New via Paid · ROAS	On-site funnel (sessions → cart → orders) · spend & blended CAC by month (combo) · new customers by acquisition channel (bar) · spend & ROAS by channel (combo) · campaign performance (table)
4	Website & Digital	Where does the on-site funnel leak?	Sessions · Conversion Rate % · Bounce Rate % · Add-to-Cart % · Cart Abandonment % · Revenue/Session	Sessions & conversion rate by month (combo) · sessions by device (donut) · revenue/session & conversion by channel (combo) · sessions & conversion by market (combo)
5	Logistics & Fulfillment	Are we delivering on time, and where is cash trapped?	On-Time Delivery % · Avg Delivery Days · Perfect Order % · Ship Cost/Order · Cost Recovery % · Cash Conversion Cycle	Avg delivery days & on-time % by month (combo) · carrier scorecard (combo) · working capital — DIO/DSO/DPO/CCC by month (column) · company cash flow (line)
6	CRM & Customer	Is the customer base retaining, and who churns?	Customers with Orders · Repeat Rate % · Avg Orders/Customer · CLV (Gross Profit) · Churned Customers · Gold-Tier Revenue %	New vs returning customer revenue by month (line) · customers by status (bar) · customers acquired by cohort month & channel (stacked column) · returns by reason (table)
7	Product & Inventory	Which SKUs and categories drive margin, and where does inventory sit?	Inventory Value · Inventory Turnover · Weeks of Cover · Return Rate % · In Transit · Supplier Spend	Gross margin % by category (bar) · return rate % by category (bar) · SKUs — revenue vs margin (scatter) · inventory value & weeks of cover by month (combo) · slow movers (table)

Every "big number + ratio" pair is a combo chart — magnitude as bars, ratio as a line on a secondary axis. Time axes are sorted oldest→newest.

Known lite-scope gaps (not polished)

See `07_report_guide.md`'s known polish backlog for the full list — undersized KPI cards/slicers (deliberately compact), two `T00_MANY_FIELDS` warnings on the working-capital column chart, no alt text yet, plain `card` visuals on the hero KPIs instead of `kpi` with target + trendline, and rendering not yet verified live in Desktop.